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VIRAL PATHOGENESIS

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Severe fever with thrombocytopenia syndrome virus infection induces thymic atrophy in IFNAR$^{-/-}$ mice

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ABSTRACT

Severe fever with thrombocytopenia syndrome virus (SFTSV) causes severe disease in humans, yet the pathogenesis remains poorly understood. A hallmark of fatal SFTS cases is the marked depletion of T cells. Previous studies on T cell depletion have predominantly focused on alterations in blood and peripheral lymphoid organs, while the thymus, a critical site for T cell development, has remained largely overlooked. In this study, we employed a lethal murine infection model to investigate the impact of SFTSV on thymic function. Our results revealed that SFTSV infected the thymus and induced severe cortical atrophy in type I interferon receptor-deficient (IFNAR$^{-/-}$) mice, characterized by a dramatic depletion of CD4$^+$CD8$^+$ double-positive (DP) thymocytes. Transcriptomic analysis indicated that thymic damage is likely attributable to impaired thymocyte proliferation and increased apoptosis, which may be a consequence of SFTSV-induced alterations in the thymic microenvironment. We found SFTSV-infected macrophages and dendritic cells in the thymus, which accumulated in the cortex and exhibited elevated secretion of IFN-γ, a cytokine commonly associated with acute thymic atrophy. These results demonstrate thymic atrophy caused by SFTSV infection and suggest potential therapeutic strategies for restoring thymic function and promoting T cell reconstitution.

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Severe fever with thrombocytopenia syndrome virus; thymic atrophy; double-positive (DP) cells depletion; increased apoptosis; elevated IFN-γ

Introduction

Severe fever with thrombocytopenia syndrome virus (SFTSV) is an emerging tick-borne pathogen that causes hemorrhagic fever with a case fatality rate of up to 30%, particularly among immunocompromised individuals and elderly patients [1,2]. The major clinical manifestations of SFTS include fever, thrombocytopenia, and leukopenia [1–3]. SFTSV is endemic in several East Asian countries, including China, South Korea, Japan, Vietnam, Thailand, and Malaysia, highlighting the public health concern posed by SFTSV [2].

An impaired adaptive immune response is a hallmark of fatal SFTS cases [4]. Numerous studies have demonstrated that CD4$^+$ and CD8$^+$ T cells are markedly depleted in patients with SFTS and experimental animal models [5–10]. The counts of peripheral blood T cells are inversely correlated with disease severity [5,6,11]. The mechanism of T cell depletion in SFTS remains incompletely defined. Existing studies have primarily focused on the impact of SFTSV on mature peripheral T cells, demonstrating that SFTSV can infect secondary lymphoid organs, including the spleen and lymph nodes, potentially leading to T cell destruction within these tissues [10,12–14]. Elevated T cell apoptosis is identified as one key contributor to this loss [8]. However, it remains unclear whether SFTSV infection may also target the thymus and consequently impairs the development of immature T cells.

The thymus, a primary lymphoid organ, serves as the central site for T cell development and differentiation. Within the thymus, bone marrow-derived lymphoid progenitor cells progress from CD4$^-$CD8$^-$ double-negative (DN) cells to CD4$^+$CD8$^+$ double-positive (DP) cells and ultimately mature into CD4$^+$CD8$^-$ or CD4$^-$CD8$^+$ single-positive (SP) T cells [15,16]. T cell differentiation occurs as cells migrate through distinct thymic regions: DN and DP cells are primarily located in the cortex, whereas mature SP cells are predominantly found in the medulla [17]. This process is orchestrated by stromal cells within the thymic microenvironment, including epithelial cells, macrophages, dendritic cells, etc. These cells regulate thymocyte proliferation and differentiation via direct cell – cell contact or secretion of cytokines such as IL-1, IL-2, IL-6, and IFN-γ [17,18].

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Studies have demonstrated that viral infections can lead to severe thymic atrophy through direct cytopathic effects or indirect disruption of the thymic microenvironment, as observed in human immunodeficiency virus (HIV), influenza virus, measles virus, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infections [19–24]. For instance, the HIV-1 Tat protein directly perturbs the DN4-to-DP transition, resulting in a marked depletion of both DP and SP thymocytes [21]. Concurrently, HIV-1 infection also elevates intrathymic IFN-β and TNF-α levels, which upregulates sphingosine-1-phosphate receptors and induces the premature export of immature DP cells [25]. Similarly, severe influenza virus infection also triggers thymic atrophy and disrupts T cell development, primarily through activation of innate $CD8^{+}CD44^{high}$ T cells that produce excessive IFN-γ [19]. Similar to SFTS, SARS-CoV-2 exhibits increased severity and mortality in elderly individuals, which is thought to be linked to age-related decline in thymic activity [16,26,27], and preservation of thymic function has been proposed as a potential strategy to improve outcomes [27–29].

Building on this evidence, we hypothesized that SFTSV infection may likewise induce thymic atrophy and disrupt T-cell development, leading to lymphopenia that critically contributes to severe SFTS. To test this, we utilized a lethal murine model and demonstrated that SFTSV directly infects the thymus, inducing severe thymic atrophy and depletion of DP thymocytes. Further mechanistic insights were pursued through transcriptomic analysis, flow cytometry, and molecular biology methods.

Materials and methods

Cells and virus

African green monkey kidney cells (Vero, ATCC CCL-81) were cultured in Dulbecco's modified Eagle's medium (DMEM, Gibco) supplemented with 10% fetal bovine serum (FBS, Gibco) at 37°C with 5% CO_2. The SFTSV-HBMC5 strain, which was isolated from a patient with SFTS in Macheng, Hubei Province, China, was used in this study [30]. Vero cells at passage 5 were infected with SFTSV at a multiplicity of infection (MOI) of 0.01 under biosafety level 2 (BSL2) conditions. The cells were then cultured in DMEM containing 2% FBS for 72 h. The viral stock was harvested and stored at −80°C until use. The virus titer was determined using an endpoint dilution assay and calculated by the Reed-Muench method as described previously [31].

Animal experiments

The animal experiments were conducted in two independent trials. The sample size of this study was determined based on relevant peer-reviewed literature related to SFTSV animal studies [31]. For hematoxylin and eosin staining (H&E), immunofluorescence assay (IFA), quantitative RT-PCR, steroid hormone detection and transcriptomics analysis, we conducted the first animal experiment to collect thymus tissues and blood specimens. 10 female 11–14-week-old type I interferon receptor-deficient ($IFNAR^{-/-}$) C57BL/6J mice were randomly assigned into two groups and intraperitoneally administered with either 10 $TCID_{50}$ (50% tissue culture infectious dose), equivalent to $100 \times 50\%$ lethal dose (LD_{50}) of SFTSV [31], or an equal volume of PBS as a control ($n = 5$ per group). The mice were euthanized at 4 days post-infection (d.p.i.). Blood samples were collected for steroid hormone detection. Thymic tissues were harvested for H&E, IFA, RT-PCR, and transcriptomics analysis. For thymocyte counting and flow cytometry analysis, we conducted the second animal experiment to harvest blood, spleen, and thymus samples. 18 female 12–17-week-old $IFNAR^{-/-}$ C57BL/6J mice were randomly assigned into three groups ($n = 6$ per group). 12 mice were infected with 10 $TCID_{50}$ of SFTSV and euthanized at 3 or 4 d.p.i. And 6 mice injected with an equal volume of PBS as a control were euthanized at 4 d.p.i. Additionally, owing to the difficulty in collecting blood from infected mice, only 3 blood samples of 3 d.p.i. group and 4 blood samples of 4 d.p.i. group were obtained (Figure 5(E)). Mice were obtained from Institute of Laboratory Animal Science, Chinese Academy of Medical Sciences and maintained under specific-pathogen-free barrier conditions at the Animal Center of Wuhan Institute of Virology. All mice were confirmed as $IFNAR^{-/-}$ by genotyping before experiment. Mice were randomly assigned to experimental groups and housed in individually ventilated cages (IVCs) under controlled environmental conditions in BSL3 laboratory.

Mice successfully infected with SFTSV exhibited significant body weight loss. A reduction of more than 15% in body weight combined with clinical signs such as hunched posture, ruffled fur, sluggish movement, etc., was defined as the euthanasia endpoint. Following isoflurane (RWD) anesthesia, mice were euthanized by enucleation for exsanguination followed by cervical dislocation. The main experimental procedures were performed by a single operator to ensure consistency in the handling of experimental samples. All animal experimental procedures were evaluated and approved by the Institutional Animal Care and Use Committee of the Wuhan Institute of Virology (WIVAF01202201).

H&E and immunofluorescence staining

Thymus samples from SFTSV-infected and control IFNAR$^{-/-}$ mice were harvested at 4 d.p.i. ($n = 5$ per group). Following fixation in 4% paraformaldehyde (Boster) for 7 days, tissues were embedded in paraffin and sectioned. 3-μm-thick serial sections were made and used for H&E staining or IFA as described previously [32]. For IFA, we treated sections with 0.1% Sudan Black B (Sigma-Aldrich) for 20 minutes to reduce autofluorescence and incubated them with primary antibodies against various markers (CD3/CD4/CD8/CD68/CD11c/Ki67/CDK1/Caspase 1/Cleaved-Caspase 3) at 4°C overnight and a secondary antibody conjugated with Alexa Fluor (AF) 488 or 555 at room temperature for 1 h. Detailed information of the antibodies, including vendors and working concentrations, is provided in the Table S1.

TUNEL staining

To evaluate apoptosis in thymus tissues, TUNEL staining was performed on thymus sections from control and infected mice at 4 d.p.i. ($n = 5$ per group). Following deparaffinization, the slices were treated with diluted Proteinase K solution (1:10, Servicebio) at 37°C for 20 min. After washing, the sections were incubated with 50 μL Equilibration Buffer (Servicebio) for 10 min at room temperature, and then labeled with TdT incubation buffer solution (Servicebio) at 37°C for 1 h. Finally, the slices were stained with DAPI (Biyuntian) and sealed with 70% glycerin (Sinopharm Chemical Reagent) and then scanned using a panoramic scanner (3DHISTECH).

Flow cytometry

To further investigate changes in thymocyte number and phenotype, flow cytometry was performed on thymus, spleen, and blood samples collected from control and infected mice at 3 and 4 d.p.i. ($n = 6$ per group). For the spleen and thymus, single-cell suspensions were prepared by mechanically dissociating the whole spleen or thymus through 70 μm cell strainers (Absin), followed by red blood cell lysis buffer (BD Biosciences) treatment for erythrocyte removal. For peripheral blood lymphocytes, diluted mouse whole blood (1:1 in PBS) was carefully layered over room-temperature lymphocyte separation medium (Dakewei), centrifuged at 500 × g for 30 minutes without brake and the lymphocytes at the plasma-lymphocyte interface were harvested. 10^6 cells per sample were stained with FVS780 (BD Biosciences) and then labeled with different antibody cocktails against surface marker at 4°C for 20 minutes (Table S1). Following surface staining, cells were fixed and permeabilized at 4°C for 2 h using Cytofix/Cytoperm (BD Biosciences) and then stained with intracellular markers (Caspase 1/Cleaved-Caspase 3/anti-SFTSV NP/IFN-γ, seen as Table S1) at 4°C for 1 h. In addition, for SFTSV NP staining, AF488-conjugated goat anti-rabbit IgG antibody was used as a second antibody and incubated at 4°C for 1 h. The stained cells were detected by LSRFortessa Cell Analyzer (BD Biosciences). Owing to a procedural oversight during sample processing, one spleen sample from the 4 d.p.i. group designated for DP cell detection was unintentionally excluded from downstream analysis (Figure 5(E)). The data were analyzed using FlowJo_v10.8.1 software. Relevant gating strategies are shown in the Figure S1.

Quantitative RT-PCR

Total RNA was extracted from thymus tissues of control and infected mice collected at 4 d.p.i. ($n = 5$ per group). Tissues were collected in sterile 1.5 mL Eppendorf tubes containing beads, and 1 mL of TRIzol (Takara) was added before storage at − 80°C. After thawing, tissue samples were homogenized (4000 rpm, 30 s) using a Tissue Cell-destroyer (NEWZONGKE), and total RNA was extracted from the tissue homogenate according to the manufacturer's protocol. RNA was converted to cDNA using PrimeScript™ RT reagent Kit with gDNA Eraser (Takara). Quantitative PCR was performed using SYBR Premix Ex Taq (Applied Biosystems) and specific primers targeting IFN-γ, TNF-α, IL-1a and IL-1b (as shown in Table S2). To quantify viral copy number, SFTSV S gene was amplified by PCR from the cDNA template, and then cloned into pMD-18 vector (Invitrogen) and used as the plasmid standard after its identity was confirmed by sequencing as described previously [31].

Transcriptomics analysis

Transcriptome sequencing was performed at The Beijing Genomics Institute using total RNA isolated from thymus tissues of control and infected mice at 4 d.p.i. ($n = 3$ per group). First, the quality of RNA samples was analyzed using Qubit 4.0 (for concentration), Agilent 2100 (for integrity), and a spectrophotometer (for purity). Once RNA samples passed quality control assessments, sequencing libraries were prepared using the MGIEasy Fast RNA Library Prep Kit (MGI), and sequencing was performed on the DNBSEQ-T7 platform. Post-sequencing, the data were analyzed using R and RStudio. Relevant information on the number of reads is shown in Table S3. We started with calculating the Pearson correlation coefficient for correlation analysis. Then, we employed DESeq2 to identify differentially expressed genes (DEGs), defined as those with $|\log2 \text{ fold change}| \geq 1$ and $Padj < 0.05$. Subsequently, KEGG enrichment analysis of the DEGs was performed. Gene set enrichment analysis (GSEA) was performed on all genes from the DESeq2 results, ranked according to log2 fold change, using the clusterProfiler R package. Heatmaps were generated to visualize the expression patterns of top 30 significant DEGs in the selected pathways.

Steroid hormone detection

To quantify hormone concentrations in the blood, liquid chromatography-tandem mass spectrometry (LC-MS) was conducted on peripheral blood samples from control and infected mice collected at 4 d.p.i. ($n = 5$ per group). The peripheral blood was centrifuged at 3000 rpm at 4°C for 15 minutes to isolate the serum. 200 µL of serum and 300 µL of methanol (Sinopharm Chemical Reagent) were mixed, vortexed, and then centrifuged at 14,000 rpm for 10 minutes. Subsequent extraction was performed on a solid-phase extraction plate and chromatographic separation was carried out using Ultra Performance Liquid Chromatography (Waters ACQUITY UPLC I-CLASS). Finally, mass spectrometry (Waters XEVO TQ-XS) was performed. The standard curve method was established to obtain quantitative results.

Statistical analysis

Data were analyzed using GraphPad Prism 9.5 software and presented as mean ± standard deviation (SD). For continuous data that following a normal distribution, differences between two groups were analyzed using an unpaired two-tailed Student's *t*-test. For comparisons among three or more groups, ordinary one-way ANOVA was used to assess overall differences between groups, followed by Dunnett's test for comparisons between infected and control groups. Data that did not follow a normal distribution were analyzed using Mann–Whitney *U* test to compare differences between the two groups (Figure 3(B)). $p < 0.05$ was considered statistically significant. *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Results

SFTSV infection caused severe thymic atrophy in IFNAR$^{-/-}$ mice

To determine whether severe SFTSV infection causes thymic damage, we infected C57BL/6J IFNAR$^{-/-}$ mice with 100× LD_{50} of SFTSV. In all the infected mice, we observed severe thymic atrophy at 3 and 4 d.p.i. (Figure 1(A)), with thymic weights reduced by 43.1% and 37.6%, respectively (Figure 1(B)). Thymocyte counts also decreased, with 60.6% and 82.8% loss at 3 and 4 d.p.i., respectively (Figure 1(C)). H&E staining showed obvious depletion of thymocytes in the thymic cortex. Uninfected mice exhibited a thick, densely packed cortex and a distinct boundary between cortex and medulla, whereas infected mice showed a reduced cortex thickness and loss of cortical thymocytes (Figure 1(D)).

Flow cytometry revealed significant decreases in counts of lymphocytes, NK cells, and dendritic cells in the infected thymus, with lymphocytes being most affected, losing 65.0% and 84.6% at 3 and 4 d.p.i., respectively (Figure 1(E)). In contrast, monocytes and macrophages showed no significant changes in the thymus (Figure 1(E)). IFA also confirmed a marked reduction in CD3$^+$, CD4$^+$, and CD8$^+$ T cells in the infected thymus, especially in the cortex (Figure 1(F)). These results suggest that SFTSV infection causes severe thymic atrophy in the cortical area and substantial depletion of lymphocytes.

SFTSV infection caused dramatic depletion of D cells

Thymus is the place of T lymphocyte development. To elucidate whether SFTSV infection affects T cell development, we employed flow cytometry with markers CD8 and CD4 to analyze four key T cell subsets: DN, DP, CD4$^+$ SP, and CD8$^+$ SP cells (Figure 2(A)). In the thymus of uninfected mice, DP cells were the most prevalent (78.0% ± 3.5%), followed by CD4$^+$ SP (13.8% ± 1.8%), DN (4.3% ± 1.1%), and CD8$^+$ SP cells (3.6% ± 1.0%) (Figure 2(B)). SFTSV infection significantly altered the proportional distribution of thymocytes, particularly reducing DP cells while increasing CD4$^+$ SP and CD8$^+$ SP cells (Figure 2(A,B)). Notably, DP cells dropped from 78.0% ± 3.5% to 53.3% ± 15.7% at 3 d.p.i. and 12.9% ± 7.9% at 4 d.p.i. (Figure 2(B)), and DP cell counts plummeted from 2.2×10^7 to 0.6×10^6 cells at 4 d.p.i. Cell counts of DN cells also declined from 1.2×10^6 to 0.8×10^5 cells, while the numbers of CD4$^+$ SP and CD8$^+$ SP

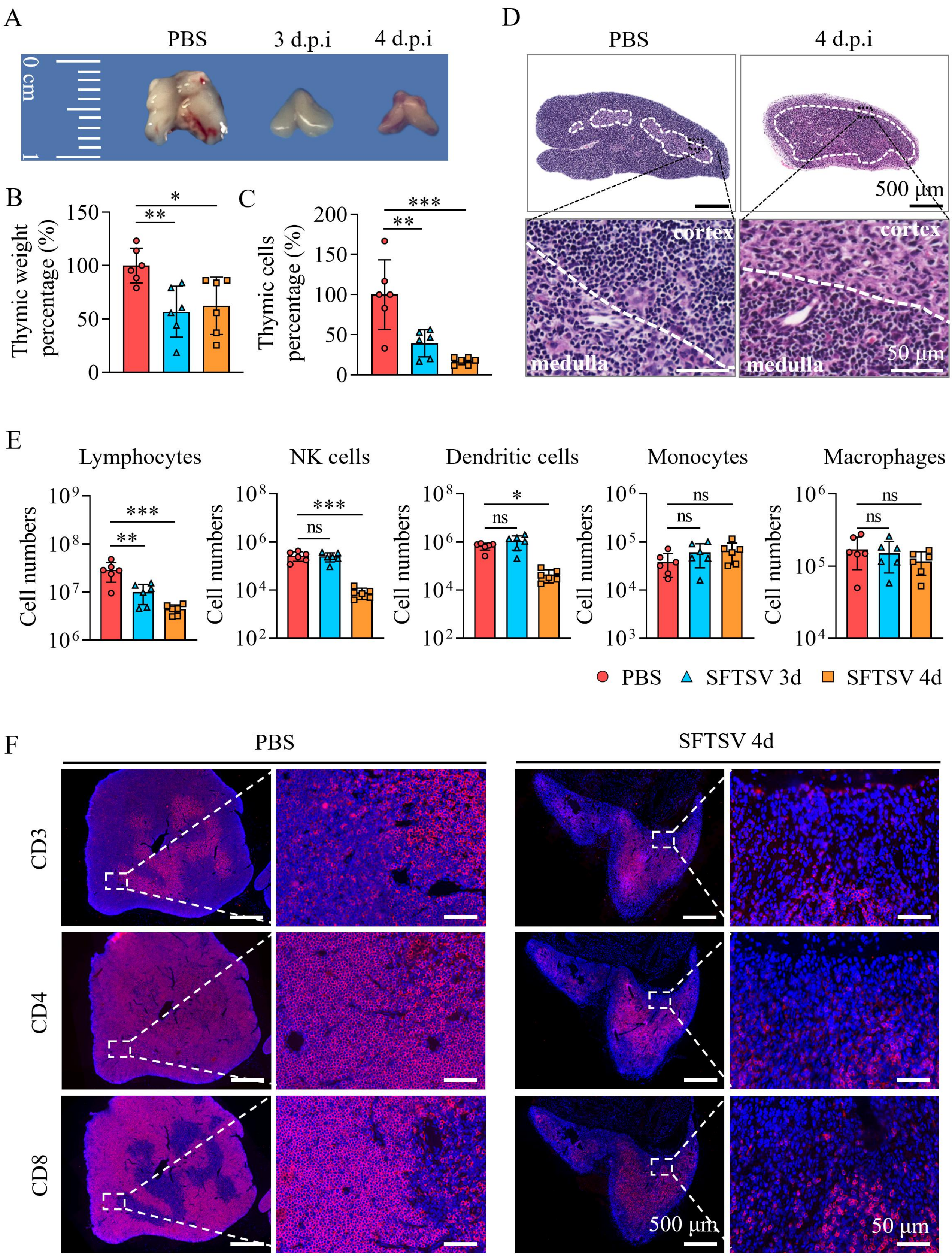

Figure 1. SFTSV induces severe thymic atrophy in IFNAR$^{-/-}$ mice. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 3 or 4 d.p.i. (A) Representative thymic morphology from control and SFTSV-infected mice. (B) Thymic weight percentage in infected mice compared with control. (C) Changes of thymocyte percentage in infected mice compared with control. (D) H&E staining of thymus in control and infected mice at 4 d.p.i. (E) Counts of thymic lymphocytes, NK cells, dendritic cells, monocytes, and macrophages, determined by flow cytometry. (F) IFA analysis of T cell subsets in the thymus of control and infected mice at 4 d.p.i. Representative data of each group are shown in (A), (D), and (F). Statistical data were showed as mean ± SD. Statistical significance was determined by one-way ANOVA ($n = 6$ per group). ns indicates $p > 0.05$, *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

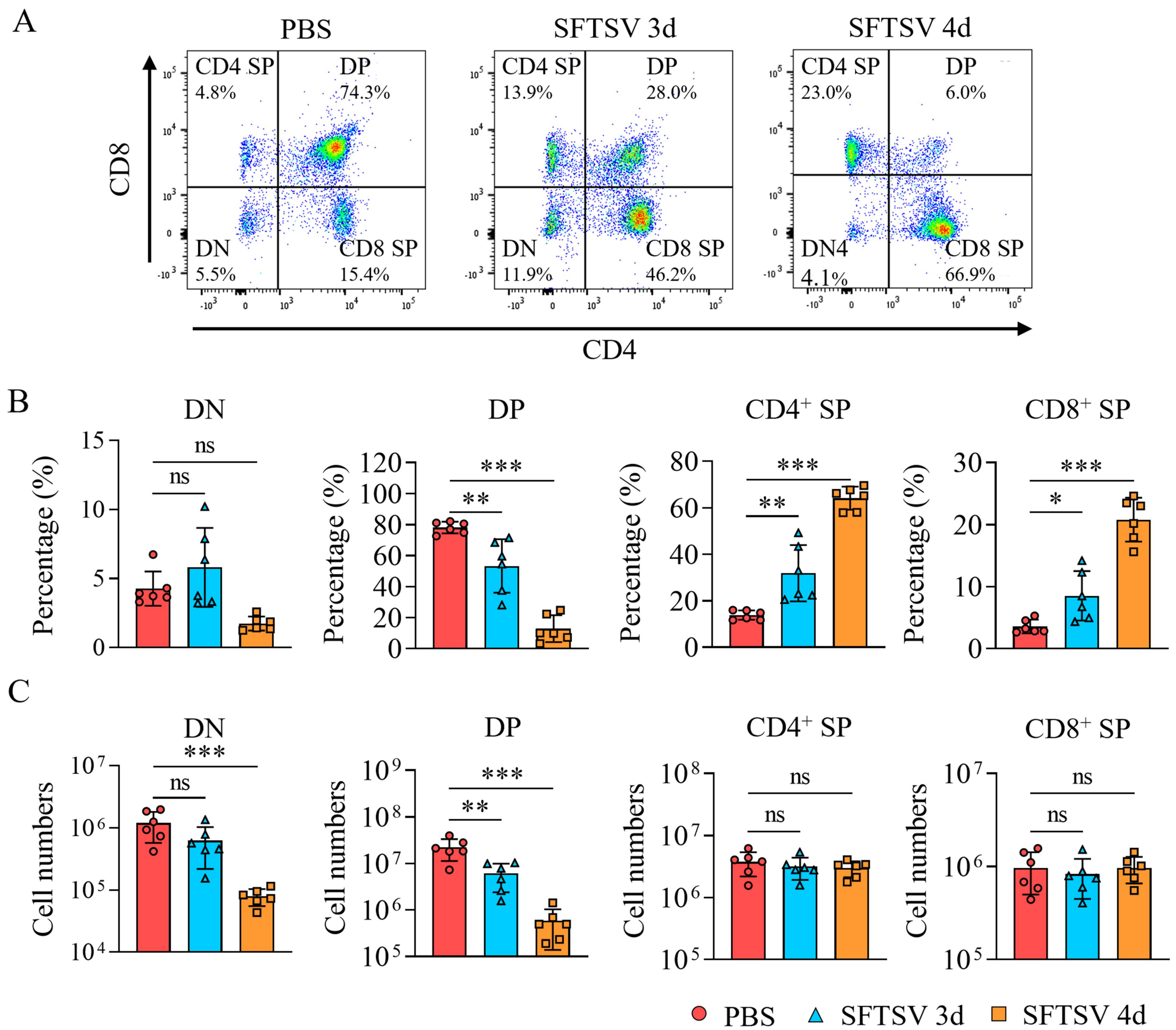

Figure 2. SFTSV infection leads to dramatic depletion of DP cells in the thymus. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 3 or 4 d.p.i. Flow cytometry was then used to analyze thymic T cell subsets gated on FVS780^{-} CD45^{+} thymic cells. (A) Changes in the ratio of thymic T cell subpopulations (DN/DP/CD4^{+} SP/CD8^{+} SP) in control and infected mice. Representative data of each group are shown. (B) Percentages and (C) absolute counts of DN, DP, and SP T cell subsets in the thymus of control and infected mice. Data were presented as mean ± SD. Statistical significance was determined by one-way ANOVA ($n = 6$ per group). ns indicates $p > 0.05$, * and ** indicate $p < 0.05$ and $p < 0.01$, respectively.

cells remained largely unchanged (Figure 2(C)). These findings indicate that SFTSV infection disrupts T cell development in the thymus, with a significant depletion of DP cells.

SFTSV mainly infected nonlymphoid cells in the thymus

To investigate whether T cell depletion was directly induced by SFTSV infection, we assessed the infection of SFTSV in the thymus. Using an anti-NP antibody, we found that at 4 d.p.i., viral antigens were present in the thymus, especially in the cortical area (Figure 3(A)). Viral RNA quantification confirmed SFTSV replication in the thymus (Figure 3(B)). Flow cytometry analysis revealed that approximately 0.9% of thymic cells were infected, mainly including macrophages (53.6% ± 9.9%) and dendritic cells (16.2% ± 3.3%) (Figure 3(C)).

To confirm the types of cells infected by SFTSV, dual immunofluorescence staining was conducted with anti-NP antibody and cellular markers. The results showed that viral NP protein largely co-localized with CD68^{+} macrophages and slightly with CD11c^{+} dendritic cells, but lymphocytes marked by CD3 were rarely infected by SFTSV (Figure 3(D)). This suggests that the significant reduction in DP cells is not due to direct infection of SFTSV.

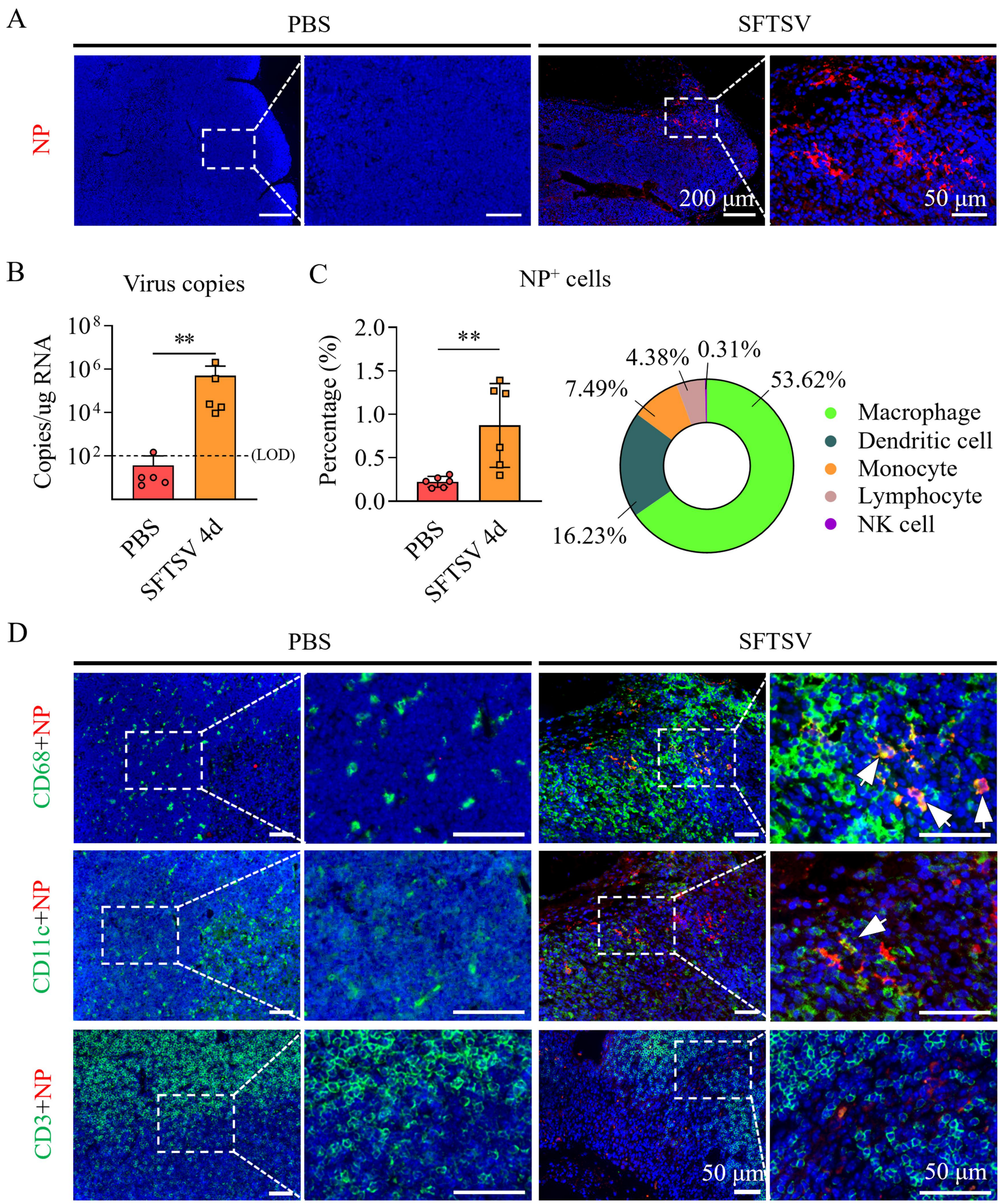

Figure 3. SFTSV mainly infects thymic macrophages and dendritic cells. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 4 d.p.i. (A) Detection of SFTSV NP protein in thymus by IFA. (B) Viral load in thymus quantified by RT-PCR targeting SFTSV S gene ($n = 5$ per group). Limit of detection (LOD) is 10^2. (C) Flow cytometry analysis of SFTSV-infected thymic cells. Left: percentage of SFTSV NP$^+$ cells gated on FVS780$^-$ thymocytes; right: composition of NP$^+$ cells in the indicated thymic subsets ($n = 6$ per group). (D) Dual-immunofluorescence staining of anti-SFTSV NP (green) and lineage cell markers (red) in the thymus of control and infected mice. CD68, CD11c, and CD3 were used for labeling macrophages, dendritic cells, and T cells, respectively. Co-localization appears yellow. Statistical data were presented as mean ± SD. Representative data of each group are shown in (A) and (D). Statistical significance was determined by Student's *t*-test (C) or Mann-Whitney *U* test (B). ns indicates $p > 0.05$, *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Cell cycle and cell death pathways were regulated in infected thymus

To uncover the mechanisms behind thymic atrophy, transcriptome analysis was conducted on the thymus of the infected and control groups. The results showed that the infected and the control groups were well separated (Figure S2A) and a total of 3,284 DEGs were identified (Figure S2B). These DEGs were involved in pathways related to viral infections, immune response, signal transduction, and cell growth and death (Figure 4(A)). A further investigation of DEGs in the cell growth and death pathways showed significant alterations in cell cycle, apoptosis, and p53 signaling pathways (Figure 4(B)). GSEA revealed significant enrichment of apoptosis (NES: 1.76),

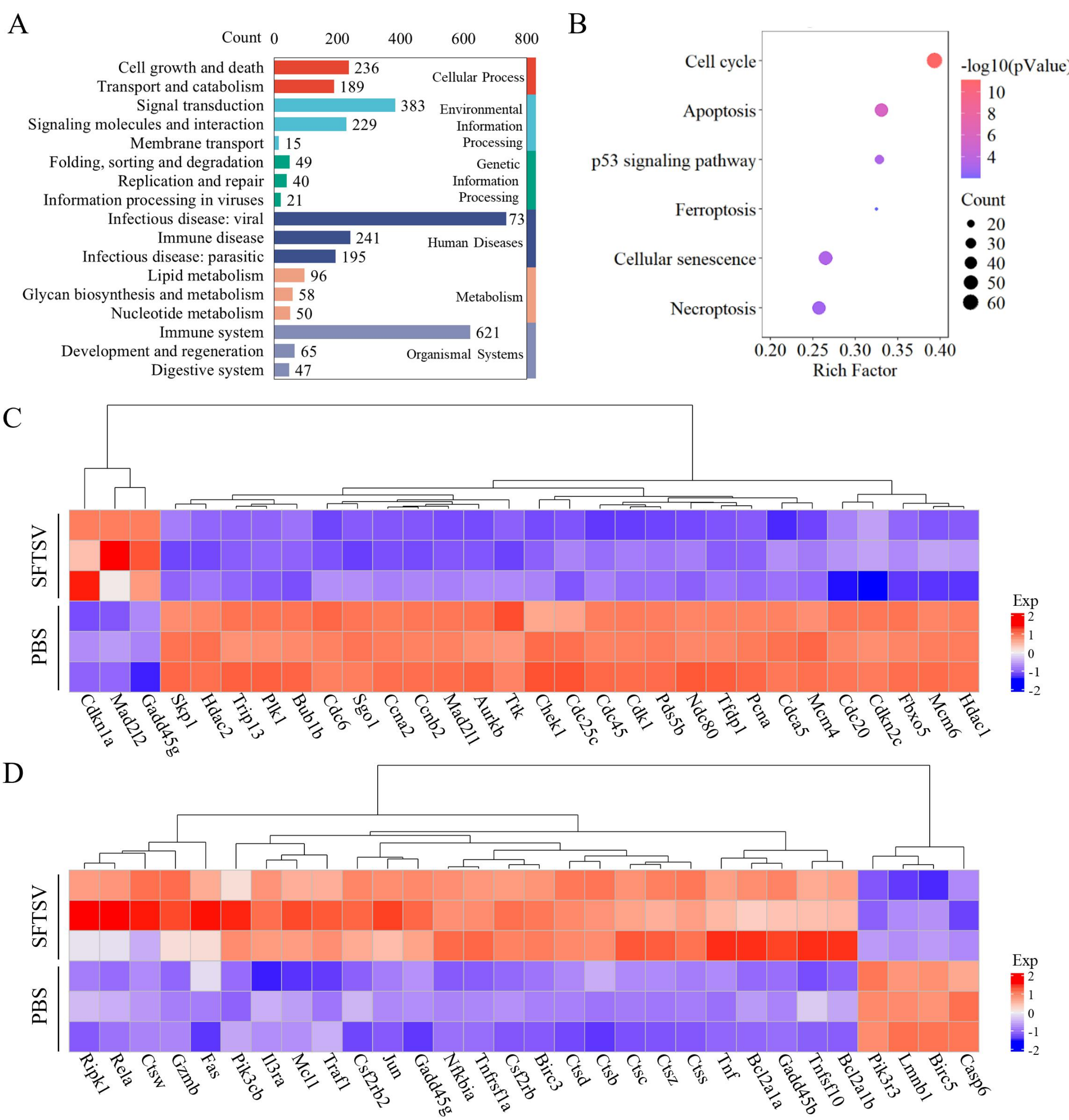

Figure 4. Transcriptomic analysis shows SFTSV-mediated regulation of cell cycle and apoptosis in thymocytes. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 4 d.p.i. Then total RNA was extracted from the thymus and RNA-seq was performed ($n = 3$ per group). (A) KEGG pathway enrichment analysis of thymic DEGs between infected and control mice. (B) Pathways associated with cell growth and death identified by KEGG analysis. (C, D) Heatmaps display the top 30 significant DEGs enriched in the cell cycle (C) and apoptosis pathways (D).

ferroptosis (NES: 1.83), and necroptosis (NES: 1.49) pathways, along with marked suppression of cell cycle-related gene sets (NES: −2.02), suggesting coordinated activation of programmed cell death coupled with cell cycle arrest (Figure S2C). The heat map of cell cycle genes revealed significant downregulation in essential mitotic factors like cyclins (Ccns), cyclin-dependent kinases (Cdks), and cell division cycle (Cdc) proteins, such as Ccna/b, Cdk1, and Cdc6/20/25/45, and upregulation in cyclin-dependent kinase inhibitors like Gadd45 and Cdkn1a (p21) after SFTSV infection (Figure 4(C)). The heat map of apoptotic genes revealed dramatic upregulation of pro-apoptotic genes such as Fas, Gzmb, Ctsb, and Prf1, as well as downregulation of anti-apoptotic genes including Pik3r3 and Birc5 (Figure 4(D)).

IFA confirmed the reduced expression of cell cycle and proliferation markers CDK1 and Ki67 in the thymus, while TUNEL staining revealed increased apoptosis in the thymic cortex (Figure S2D). These findings demonstrate reduced thymocyte proliferation and increased apoptosis following SFTSV infection, which may contribute to SFTSV-induced thymic atrophy.

Apoptosis appeared to be one of the main reasons for dp depletion

TUNEL staining revealed that apoptotic cells in the infected thymus were mainly in the cortex, consistent with the significant loss of DP cells, which were typically found in cortical zones. Flow cytometry analysis using Apotracker, a probe for detecting apoptotic cells, showed an increase in apoptotic thymocytes from 2.2% ± 1.5% in controls to 14.7% ± 8.1% at 3 d.p.i. and 17.1% ± 3.6% at 4 d.p.i (Figure 5(A)). Over 32% of these Apotracker$^+$ thymocytes were DP cells at 3 d.p.i., followed by CD4$^+$ SP (26.4% ± 7.4%) and CD8$^+$ SP cells (7.1% ± 2.7%) (Figure 5(B)). The percentage of apoptotic DP cells rose from 1.5% ± 0.9% in uninfected mice to 13.7% ± 6.2% at 3 d.p.i. and 16.1% ± 6.6% at 4 d.p.i (Figure 5(C)). The mean fluorescence intensity (MFI) of Apotracker (from 224.7 to 983.7 and 1632.3) and Cleaved-Caspase 3 (from 195.7 to 346.0 and 573.0) in DP cells significantly increased, indicating heightened apoptosis (Figure 5(D), left and middle panels). Although MFI of Caspase 1, associated with pyroptosis, also showed an increase (from 34.9 to 57.4 and 132.5), IFA revealed significantly less pronounced upregulation compared to Cleaved-Caspase 3 (Figure 5(D), right panel, and Figure S3). Therefore, it appears that both apoptosis and pyroptosis contribute to DP cell depletion, with apoptosis being more predominant.

We also checked if migration of immature DP cells from the thymus to peripheral organs might be another cause for the DP cell decline. No significant increase of DP cells was found in peripheral blood or spleen (Figure 5(E)), indicating that premature migration of DP cells was not a cause of thymic atrophy. In conclusion, apoptosis, rather than premature migration, appears to be the primary cause of DP cell depletion.

Infected macrophages and dendritic cells exhibited enhanced IFN-γ secretion

Hormones and some cytokines (like IFN-γ, TNF-α, and IL-1) secreted by thymic microenvironment cells, such as dendritic cells, macrophages, thymic epithelial cells, and mesenchymal stromal cells etc., play a crucial role in T cell differentiation. The serum levels of corticosterone (from 359.4 ng/ml to 472.6 ng/ml) and testosterone (from 0.05 ng/ml to 0.04 ng/ml) were determined, and no significant increase was found in infected mice (Figure 6(A)). The cytokines associated with thymic atrophy were analyzed using RT-qPCR, and a significant upregulation of IFN-γ, TNF-α, IL-1a, and IL-1b in the thymus of SFTSV-infected mice was identified (Figure 6(B)). The expression of IFN-γ was further assessed using flow cytometry, which showed a time-dependent increase at 3 and 4 d.p.i. (from 41.9 to 77.2 and 146.0, Figure 6(C)). Further analysis showed that expression of IFN-γ was significantly elevated across multiple thymic immune subsets, including macrophages (from 604.7 to 702.3 and 882.5), dendritic cells (from 150.3 to 132.0 and 456.2), NK cells (from 160.3 to 200.3 and 186.5), lymphocytes (from 103.5 to 122.5 and 158.2), etc (Figure 6(D)). And infected macrophages (from 296.5 to 681.7) and dendritic cells (from 76.6 to 260.0) showed a substantial increase in IFN-γ expression at 4 d.p.i (Figure 6(E)). These data suggest that SFTSV infection alters the thymic microenvironment, and infected macrophages and dendritic cells exhibited enhanced IFN-γ secretion.

Interestingly, although macrophage and dendritic cell numbers in the thymus did not increase (Figure 1 (E)), their distribution changed, with a more concentrated distribution in the cortex of infected thymus compared with a scattered distribution in controls (Figure 6(F)).

Discussion

Previous immunological studies on SFTS have primarily focused on the effector immune response and alterations in blood and peripheral lymphoid organs, with limited attention given to the thymus [5–14]. Our study

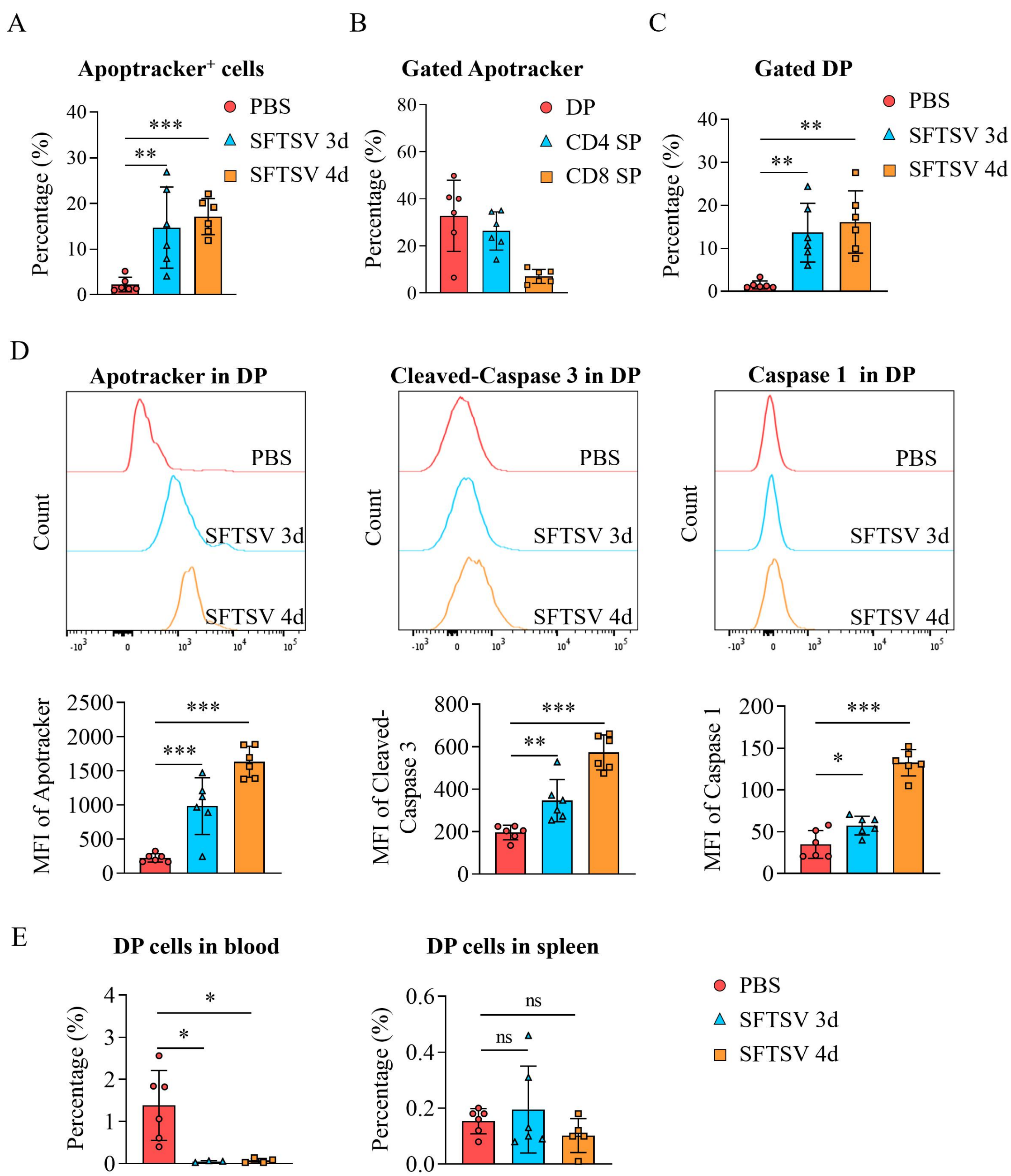

Figure 5. SFTSV infection causes obvious apoptosis of DP cells but not premature migration. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 3 or 4 d.p.i. Flow cytometry was then used to analyze apoptosis in thymic T cells. (A) Proportion of thymic apoptotic cells (Apotracker$^+$) in control and infected mice ($n = 6$ per group). The FACS plots were gated on thymocytes. (B) Proportions of DP, CD4$^+$ SP and CD8$^+$ SP cells among apoptotic cells (Apotracker$^+$) in control and infected mice at 3 d.p.i. ($n = 6$ per group). (C) Proportions of apoptotic cells among DP cells in control and infected mice ($n = 6$ per group). (D) Expression levels of Apotracker/Cleaved-Caspase 3/Caspase 1 among DP cells in control and infected mice ($n =$ 6 per group). Representative data of each group are shown. (E) Frequency of DP cells among lymphocytes in peripheral blood and spleen of control and infected mice ($n = 3$, 4, or 6 per group). Cells were gated on the lymphocyte population ($FSC^{low}SSC^{low}$). Statistical data were presented as mean ± SD. Statistical significance was determined by one-way ANOVA. ns indicates $p > 0.05$. *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

demonstrated that SFTSV infection led to severe thymic atrophy marked by a significant loss of lymphocytes, particularly DP cells (Figure 7(A)). This damage, manifested as reduced thymocyte proliferation and increased apoptosis, is likely attributable to SFTSV-induced alterations in the thymic microenvironment

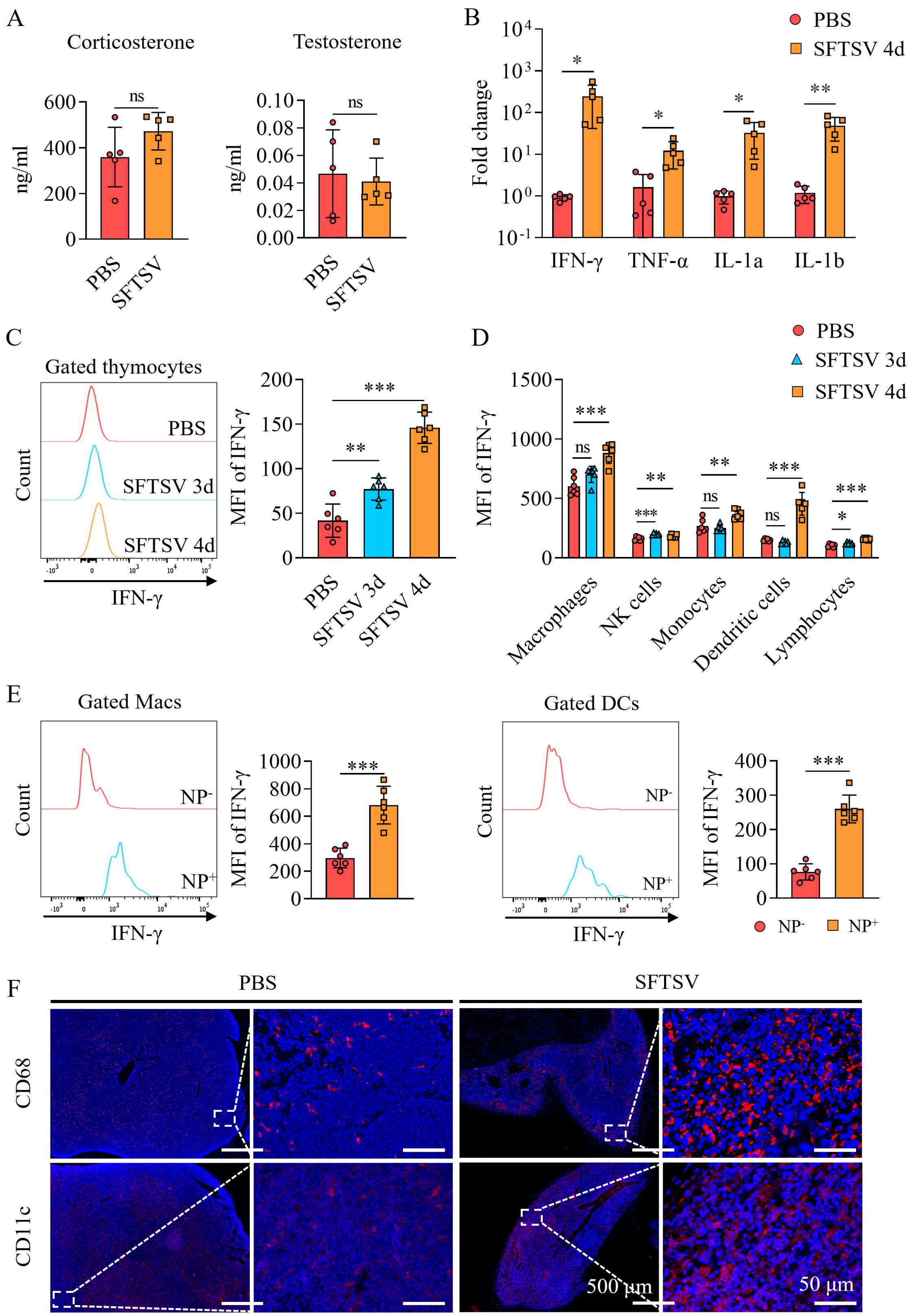

Figure 6. SFTSV infection alters the thymic microenvironment. IFNAR$^{-/-}$ C57BL/6J mice were infected intraperitoneally with 10 $TCID_{50}$ of SFTSV or PBS (control) and euthanized at 3 or 4 d.p.i. (A) Serum corticosterone and testosterone concentrations in control and infected mice at 4 d.p.i. ($n = 5$ per group). (B) RT-qPCR analysis of cytokine mRNA expression of thymus in control and infected mice at 4 d.p.i. ($n = 5$ per group). (C) Intracellular IFN-γ expression in the thymus of control and infected mice analyzed by flow cytometry ($n = 6$ per group). FACS plots were gated on FVS780^{-} thymocytes. (D) IFN-γ expression in indicated thymic subsets of control and infected mice gated on macrophages, NK cells, monocytes, dendritic cells, and lymphocytes ($n = 6$ per group). (E) IFN-γ

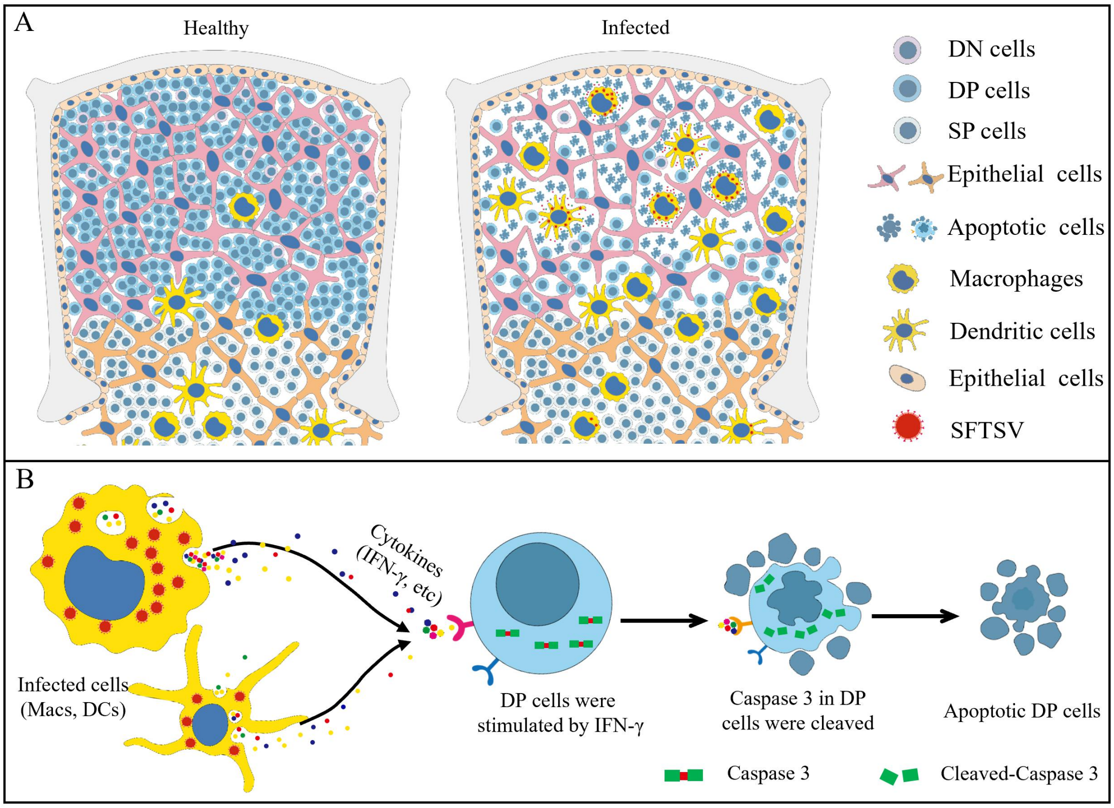

Figure 7. Schematic model of thymic involution caused by SFTSV infection. (A) SFTSV could infect the macrophages and dendritic cells in the thymus and led to dramatic loss of DP cells accompanied by aggregation of macrophages and dendritic cells in the cortex. (B) The infected macrophages and dendritic cells produced massive INF-γ, which might induce obvious apoptosis activation of the DP cells.

rather than direct viral cytopathic effects. Our results suggest that SFTSV-infected macrophages and dendritic cells, which accumulate in the cortical region of the thymus, secrete elevated levels of IFN-γ, thereby being involved in triggering apoptosis and depletion of DP cells, ultimately leading to thymic atrophy (Figure 7(B)).

Despite the presence of the blood-thymus barrier, various viruses have been detected in the thymus, where they target distinct cell populations and can induce thymic atrophy. For example, HIV infects $FoxP3^{+}CD3^{high}CD8^{-}$ thymic lymphocytes directly [33], porcine reproductive and respiratory syndrome virus (PRRSV) targets thymic $CD14^{+}$ monocytes and macrophages [34], influenza A virus is detected in thymic dendritic cells [19], and SARS-CoV-2 infects thymic epithelial cells and thymocytes [35,36]. Interestingly, a common feature of severe thymic atrophy is the decline in DP cells, which are usually not directly infected by these viruses. This may be because the DP cells exhibit heightened sensitivity to apoptosis induced by certain factors compared with SP cells [37]. Our study revealed that SFTSV infection induces severe thymic atrophy with a pronounced loss of DP thymocytes. Paradoxically, the viral antigen was not detected in these diminishing cells but was instead found primarily in macrophages and dendritic cells (Figure 3). These findings suggest that SFTSV-induced thymic atrophy occurs via an indirect mechanism, in which infected macrophages and dendritic cells play a contributory role, a pattern also observed with other viruses.

Thymic atrophy in the context of infectious diseases is typically triggered by several key factors: impaired

expression in uninfected (NP^{-}) versus infected (NP^{+}) macrophages (Macs) and dendritic cells (DCs) at 4 d.p.i. analyzed by flow cytometry ($n = 6$ per group). (F) IFA staining of macrophages (CD68) and dendritic cells (CD11c) in thymus of control and infected mice at 4 d.p.i. Representative data of each group are shown in (C), (E), and (F). Data were presented as mean ± SD. Statistical significance was determined by Student's *t*-test. ns indicates $p > 0.05$. *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

cell proliferation, increased cell death, and premature migration of thymocytes to peripheral lymphoid tissues [19]. In parasitic infections, thymic involution is commonly associated with reduced proliferation and premature egress. A key example is *Trypanosoma cruzi* infection, which disrupts S1P-related signaling to drive the premature exit of immature thymocytes [38]. Differently, viral infections – such as those caused by influenza virus, PRRSV, chicken anemia virus, and mouse hepatitis virus – more frequently induce thymic atrophy through elevated thymocyte death [19,39–41]. In our study, we also did not observe evidence of premature DP cell migration to peripheral tissues (Figure 5). Instead, SFTSV infection induced cell cycle arrest and significant apoptosis in thymocytes (Figures 4 and 5). These findings suggest that both suppression of cell proliferation and enhanced apoptosis play critical roles in SFTSV-induced thymic atrophy.

Although the precise mechanisms of acute infection-induced thymic atrophy remain incompletely defined, hormones (including glucocorticoids and testosterone) and pro-inflammatory mediators such as IFN-γ, TNF-α, and IL-6 are recognized as established contributors [19,42–44]. In our study, we found no change in testosterone and only a non-significant increase in corticosterone during SFTSV infection, suggesting that classic stress hormones are not the primary drivers of thymic atrophy in this model. Instead, we identified a pronounced induction of IFN-γ, predominantly produced by infected macrophages and dendritic cells (Figure 6(B-E)). Notably, these IFN-γ–producing cells accumulated in the thymic cortex – the region densely populated with DP thymocytes. This spatial redistribution likely establishes a localized microenvironment of high IFN-γ concentration, which we propose drives the observed DP cell apoptosis and suppression of proliferation [45]. This hypothesis is consistent with studies in influenza and SARS-CoV-2 infections, where IFN-γ neutralization has been shown to rescue DP cell loss [19,36].

A confusing issue is whether viral NP in thymic myeloid cells originates from active replication or phagocytic clearance. Our results revealed that myeloid cells constituted the vast majority (77.3%) of NP^+ cells at 4 d.p.i., whereas infection in other thymic populations was rarely observed (Figure 3(C–D)). This distribution strongly argues against phagocytosis as the major source of NP. Instead, our findings support a model in which SFTSV directly infects thymic macrophages and dendritic cells, as observed in other studies [13,46,47]. Following infection, these cells not only amplify the virus but also produce inflammatory mediators such as IFN-γ, creating a localized inflammatory milieu that induces bystander apoptosis and impairs DP thymocyte proliferation, thereby exacerbating thymic atrophy.

This study has several limitations. Our conclusions are restricted to a murine model, specifically relying on $IFNAR^{-/-}$ mice due to the resistance of wild-type animals to SFTSV. This model carries an inherent limitation, as thymic development and immunity in this model are fundamentally shaped by the absence of IFN-I responses [48]. Furthermore, the rapid disease course in this model prevented us from studying potential thymic recovery after the initial injury. Finally, the critical role we propose for IFN-γ was not verified by *in vivo* neutralization experiments. Future work should prioritize developing immunocompetent murine models susceptible to SFTSV to validate the mechanisms of thymic atrophy and assess the therapeutic potential of targeting IFN-γ.

In conclusion, our study reveals severe thymic atrophy in SFTSV-infected $IFNAR^{-/-}$ mice. We found that SFTSV infects macrophages and dendritic cells within the thymus, enhanced their secretion of IFN-γ, and alters their distribution. These changes may lead to suppressed thymocyte proliferation and increased apoptosis, ultimately resulting in thymic atrophy. Thymic damage likely impairs the immune system in clearing the virus. Our findings shed light on the pathogenic mechanisms of SFTSV and suggest potential therapeutic approaches that aim at restoring thymic function for patients with severe SFTS.

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Author contributions

CRediT: **Longda Ma:** Data curation, Formal analysis, Methodology, Writing – original draft; **Hui Zhang:** Data curation; **Manli Wang:** Conceptualization, Resources, Writing – review & editing; **Zhihong Hu:** Conceptualization, Resources, Writing – review & editing; **Yiwu Zhou:** Resources, Writing – review & editing; **Jia Liu:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Statement of adherence to arrive guidelines

We have adhered to ARRIVE guidelines. The complete ARRIVE guidelines checklist was included in S1 The ARRIVE Guidelines checklist.

Data availability statement

The data supporting the findings of this study are available in Figshare at DOI: https://doi.org/10.6084/m9.figshare.29364545.v1 [49].

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References

[1] Chen L, Chen T, Li R, et al. Recent advances in the study of the immune escape mechanism of SFTSV and its therapeutic agents. Viruses. 2023;15(4):940. doi: 10.3390/v15040940
[2] Kim D, Lai CJ, Cha I, et al. Current progress of severe fever with thrombocytopenia syndrome virus (SFTSV) vaccine development. Viruses. 2024;16(1):128. doi: 10.3390/v16010128
[3] Xu S, Jiang N, Nawaz W, et al. Infection of humanized mice with a novel phlebovirus presented pathogenic features of severe fever with thrombocytopenia syndrome. PLOS Pathog. 2021;17(5):e1009587. doi: 10.1371/journal.ppat.1009587
[4] Song P, Zheng N, Liu Y, et al. Deficient humoral responses and disrupted B-cell immunity are associated with fatal SFTSV infection. Nat Commun. 2018;9(1):3328. doi: 10.1038/s41467-018-05746-9
[5] Liu J, Wang L, Feng Z, et al. Dynamic changes of laboratory parameters and peripheral blood lymphocyte subsets in severe fever with thrombocytopenia syndrome patients. Int J Infect Dis. 2017;58:45–51. doi: 10.1016/j.ijid.2017.02.017
[6] Sun L, Hu Y, Niyonsaba A, et al. Detection and evaluation of immunofunction of patients with severe fever with thrombocytopenia syndrome. Clin Exp Med. 2014;14(4):389–395. doi: 10.1007/s10238-013-0259-0
[7] Lu QB, Cui N, Hu JG, et al. Characterization of immunological responses in patients with severe fever with thrombocytopenia syndrome: a cohort study in China. Vaccine. 2015;33(10):1250–1255. doi: 10.1016/j.vaccine.2015.01.051
[8] Li MM, Zhang WJ, Liu J, et al. Dynamic changes in the immunological characteristics of T lymphocytes in surviving patients with severe fever with thrombocytopenia syndrome (SFTS). Int J Infect Dis. 2018;70:72–80. doi: 10.1016/j.ijid.2018.03.010
[9] Ji M, Hu J, Zhang D, et al. Inhibition of SFTSV replication in humanized mice by a subcutaneously administered anti-PD1 nanobody. EMBO Mol Med. 2024;16(3):575–595. doi: 10.1038/s44321-024-00026-0
[10] Sakai Y, Mura S, Kuwabara Y, et al. Lethal severe fever with thrombocytopenia syndrome virus infection causes systemic germinal centre failure and massive T cell apoptosis in cats. Front Microbiol. 2023;14:1333946. doi: 10.3389/fmicb.2023.1333946
[11] Li MM, Zhang WJ, Weng XF, et al. Cd4 T cell loss and Th2 and Th17 bias are associated with the severity of severe fever with thrombocytopenia syndrome (SFTS). Clin Immunol. 2018;195:8–17. doi: 10.1016/j.clim.2018.07.009
[12] Westover JB, Hickerson BT, Van Wettere AJ, et al. Vascular leak and hypercytokinemia associated with severe fever with thrombocytopenia syndrome virus infection in mice. Pathogens. 2019;8(4):158. doi: 10.3390/pathogens8040158
[13] Suzuki T, Sato Y, Sano K, et al. Severe fever with thrombocytopenia syndrome virus targets B cells in lethal human infections. J Clin Invest. 2020;130(2):799–812. doi: 10.1172/jci129171
[14] Takahashi T, Maeda K, Suzuki T, et al. The first identification and retrospective study of severe fever with thrombocytopenia syndrome in Japan. J Infect Dis. 2014;209(6):816–827. doi: 10.1093/infdis/jit603
[15] Thapa P, Farber DL. The role of the thymus in the immune response. Thorac Surg Clin. 2019;29(2):123–131. doi: 10.1016/j.thorsurg.2018.12.001
[16] Gruver AL, Hudson LL, Sempowski GD. Immunosenescence of ageing. J Pathol. 2007;211(2):144–156. doi: 10.1002/path.2104
[17] Savino W, Mendes-Da-Cruz DA, Smaniotto S, et al. Molecular mechanisms governing thymocyte migration: combined role of chemokines and extracellular matrix. J Leukoc Biol. 2004;75(6):951–961. doi: 10.1189/jlb.1003455
[18] Mignini F, Sabbatini M, Mattioli L, et al. Neuro-immune modulation of the thymus microenvironment (review). Int J Mol Med. 2014;33(6):1392–1400. doi: 10.3892/ijmm.2014.1709
[19] Liu B, Zhang X, Deng W, et al. Severe influenza A (H1N1)pdm09 infection induces thymic atrophy through activating innate CD8(+)CD44(hi) T cells by upregulating IFN-γ. Cell Death Dis. 2014;5(10):e1440. doi: 10.1038/cddis.2014.323
[20] Zhang JL, Li YH, Wang LL, et al. Azvudine is a thymus-homing anti-SARS-CoV-2 drug effective in treating COVID-19 patients. Signal Transduct Target Ther. 2021;6(1):414. doi: 10.1038/s41392-021-00835-6
[21] Fiume G, Scialdone A, Albano F, et al. Impairment of T cell development and acute inflammatory response in HIV-1 Tat transgenic mice. Sci Rep. 2015;5(1):13864. doi: 10.1038/srep13864

[22] Permar SR, Moss WJ, Ryon JJ, et al. Increased thymic output during acute measles virus infection. J Virol. 2003;77(14):7872–7879. doi: 10.1128/jvi.77.14.7872-7879.2003
[23] Valentin H, Azocar O, Horvat B, et al. Measles virus infection induces terminal differentiation of human thymic epithelial cells. J Virol. 1999;73(3):2212–2221. doi: 10.1128/jvi.73.3.2212-2221.1999
[24] Gonçalves R, Couto J, Ferreirinha P, et al. SARS-CoV-2 variants induce distinct disease and impact in the bone marrow and thymus of mice. iScience. 2023;26(2):105972. doi: 10.1016/j.isci.2023.105972
[25] Resop RS, Salvatore B, Kim SJ, et al. HIV-1 infection results in sphingosine-1-phosphate receptor 1 dysregulation in the human thymus. Int J Mol Sci. 2023;24(18):13865. doi: 10.3390/ijms241813865
[26] Zhou F, Yu T, Du R, et al. Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study. Lancet. 2020;395(10229):1054–1062. doi: 10.1016/s0140-6736(20)30566-3
[27] Wang W, Thomas R, Oh J, et al. Thymic aging may be associated with COVID-19 pathophysiology in the elderly. Cells. 2021;10(3):628. doi: 10.3390/cells10030628
[28] Minutolo A, Petrone V, Fanelli M, et al. Thymosin alpha 1 restores the immune homeostasis in lymphocytes during post-acute sequelae of SARS-CoV-2 infection. Int Immunopharmacol. 2023;118:110055. doi: 10.1016/j.intimp.2023.110055
[29] Liu Y, Pan Y, Hu Z, et al. Thymosin alpha 1 reduces the mortality of severe coronavirus disease 2019 by restoration of lymphocytopenia and reversion of exhausted T cells. Clin Infect Dis. 2020;71(16):2150–2157. doi: 10.1093/cid/ciaa630
[30] Zhang Y, Shen S, Shi J, et al. Isolation, characterization, and phylogenic analysis of three new severe fever with thrombocytopenia syndrome bunyavirus strains derived from Hubei Province, China. Virol Sin. 2017;32(1):89–96. doi: 10.1007/s12250-017-3953-3
[31] Wang X, Xu M, Ke H, et al. Construction and characterization of severe fever with thrombocytopenia syndrome virus with a fluorescent reporter for antiviral drug screening. Viruses. 2023;15(5):1147. doi: 10.3390/v15051147
[32] Liu J, Li Y, Liu Q, et al. SARS-CoV-2 cell tropism and multiorgan infection. Cell Discov. 2021;7(1):17. doi: 10.1038/s41421-021-00249-2
[33] Swaminathan S, Scorza T, Yero A, et al. Impact of in vitro HIV infection on human thymic regulatory T cell differentiation. Front Microbiol. 2023;14:1217801. doi: 10.3389/fmicb.2023.1217801
[34] Wang G, Yu Y, Cai X, et al. Effects of PRRSV infection on the porcine thymus. Trends Microbiol. 2020;28(3):212–223. doi: 10.1016/j.tim.2019.10.009
[35] Rosichini M, Bordoni V, Silvestris DA, et al. SARS-CoV-2 infection of thymus induces loss of function that correlates with disease severity. J Allergy Clin Immunol. 2023;151(4):911–921. doi: 10.1016/j.jaci.2023.01.022
[36] Rizvi ZA, Sadhu S, Dandotiya J, et al. SARS-CoV-2 infection induces thymic atrophy mediated by IFN-γ in hACE2 transgenic mice. Eur J Immunol. 2024;54(7):e2350624. doi: 10.1002/eji.202350624
[37] van den Brandt J, Wang D, Reichardt HM. Resistance of single-positive thymocytes to glucocorticoid-induced apoptosis is mediated by CD28 signaling. Mol Endocrinol. 2004;18(3):687–695. doi: 10.1210/me.2003-0390
[38] Lepletier A, de Almeida L, Santos L, et al. Early double-negative thymocyte export in Trypanosoma cruzi infection is restricted by sphingosine receptors and associated with human Chagas disease. PLOS Negl Trop Dis. 2014;8(10):e3203. doi: 10.1371/journal.pntd.0003203
[39] Castaño P, Benavides J, Lee MS, et al. Tissue tropism of chicken anaemia virus in naturally infected broiler chickens. J Comp Pathol. 2019;167:32–40. doi: 10.1016/j.jcpa.2018.11.008
[40] Godfraind C, Holmes KV, Coutelier JP. Thymus involution induced by mouse hepatitis virus A59 in BALB/c mice. J Virol. 1995;69(10):6541–6547. doi: 10.1128/jvi.69.10.6541-6547.1995
[41] Li Y, Wang G, Liu Y, et al. Identification of apoptotic cells in the thymus of piglets infected with highly pathogenic porcine reproductive and respiratory syndrome virus. Virus Res. 2014;189:29–33. doi: 10.1016/j.virusres.2014.04.011
[42] Deobagkar-Lele M, Chacko SK, Victor ES, et al. Interferon-γ- and glucocorticoid-mediated pathways synergize to enhance death of CD4(+) CD8(+) thymocytes during Salmonella enterica serovar Typhimurium infection. Immunology. 2013;138(4):307–321. doi: 10.1111/imm.12047
[43] Chen W, Kuolee R, Austin JW, et al. Low dose aerosol infection of mice with virulent type A Francisella tularensis induces severe thymus atrophy and CD4+CD8+ thymocyte depletion. Microb Pathog. 2005;39(5–6):189–196. doi: 10.1016/j.micpath.2005.08.005
[44] Wang S, Lyu C, Duan G, et al. Streptococcus suis serotype 2 infection causes host immunomodulation through induction of thymic atrophy. Infect Immun. 2020;88(4). doi: 10.1128/iai.00950-19
[45] Bhat MY, Solanki HS, Advani J, et al. Comprehensive network map of interferon gamma signaling. J Cell Commun Signal. 2018;12(4):745–751. doi: 10.1007/s12079-018-0486-y
[46] Matsuno K, Orba Y, Maede-White K, et al. Animal models of emerging tick-borne phleboviruses: determining target cells in a lethal model of SFTSV infection. Front Microbiol. 2017;8:104. doi: 10.3389/fmicb.2017.00104
[47] Hofmann H, Li X, Zhang X, et al. Severe fever with thrombocytopenia virus glycoproteins are targeted by neutralizing antibodies and can use DC-SIGN as a receptor for pH-dependent entry into human and animal cell lines. J Virol. 2013;87(8):4384–4394. doi: 10.1128/jvi.02628-12
[48] Martinez RJ, Hogquist KA. The role of interferon in the thymus. Curr Opin Immunol. 2023;84:102389. doi: 10.1016/j.coi.2023.102389
[49] Ma L, Zhang H, Wang M, et al. Severe fever with thrombocytopenia syndrome virus infection induces thymic atrophy in IFNAR-/- mice. Figshare Dataset. 2025. doi: 10.6084/m9.figshare.29364545.v1